Superposition of Waves: Fundamentals

General Equation of Wave Motion:

$$\frac{d^2y}{dt^2} = v^2\frac{d^2y}{dx^2}$$

The wave equation $y(x,t) = f\left(t \pm \frac{x}{v}\right)$, where $y(x,t)$ should be finite everywhere.

$f\left(t + \frac{x}{v}\right)$ represents a wave travelling in the -ve x-axis → negative direction

$f\left(t - \frac{x}{v}\right)$ represents a wave travelling in the +ve x-axis → positive direction

Terms Related to Wave Motion:

- Wave Number (or Propagation Constant): $k = \frac{2\pi}{\lambda} = \frac{\omega}{v}$
- Phase of Wave: The argument of the harmonic function $(\omega t \pm kx + \phi)$ is called the phase of the wave.
- Phase difference $(\Delta\phi)$: $\boxed{\Delta\phi = \frac{2\pi}{\lambda}\Delta x}$ where Δx is path difference.

String Dynamics:

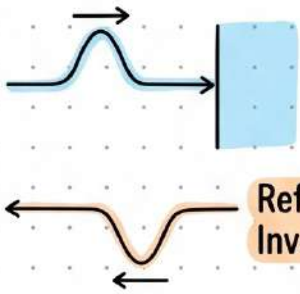
- Speed of Transverse Wave Along the String: $\boxed{v = \sqrt{\frac{T}{\mu}}}$ (where T is Tension and μ is mass per unit length).
- Average Power: $\langle P\rangle = 2\pi^2 f^2 A^2 \mu v$

Reflection of Waves:

- Equation of wave reflected at a rigid boundary:

$$y_r(x,t) = a\sin(kx + \omega t + \pi)$$

The reflected wave at a rigid boundary is 180° out of phase.

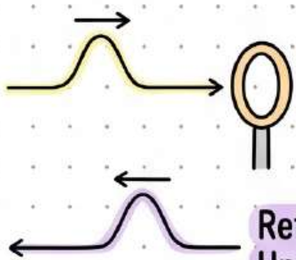


Reflected pulse: Inverted & Reversed

- Equation of wave reflected at an open boundary:

$$y_r(x,t) = a\sin(kx + \omega t)$$

The reflected wave at an open boundary is in phase with the incident wave.



Reflected pulse: Upright & Reversed

Standing Waves & Harmonics

Standing/Stationary Waves

$$y_1 + y_2 = 2A\cos\left(kx + \frac{\theta_2 - \theta_1}{2}\right)\sin\left(\omega t + \frac{\theta_1 + \theta_2}{2}\right)$$

The quantity $2A\cos\left(kx + \frac{\theta_2 - \theta_1}{2}\right)$ represents resultant amplitude at x.

- At some position resultant amplitude is zero these are called nodes, and at some positions resultant amplitude is 2A, these are called anti-nodes.
- Distance between successive nodes or anti-nodes = $\frac{\lambda}{2}$
- Distance between adjacent node and anti-node = $\lambda/4$
- All the particles in same segment (portion between two successive nodes) vibrate in same phase.
- Since nodes are permanently at rest so energy can not be transmitted across these.

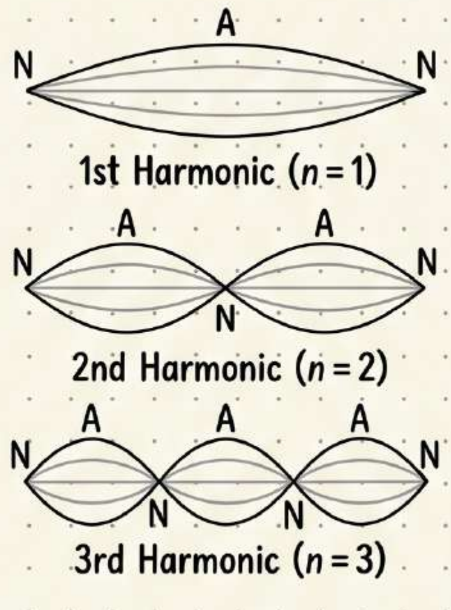
Vibrations of Strings (Fixed at Both Ends)

First harmonics or Fundamental frequency:

$$f_1 = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$$

nth harmonics or $(n-1)$th overtone:

$$f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}}$$



String Free at One End

First harmonics or Fundamental frequency: $f_1 = \frac{1}{4L}\sqrt{\frac{T}{\mu}}$

$(2n+1)$th harmonic or nth overtone: $f_{2n+1} = \frac{(2n+1)}{4L}\sqrt{\frac{T}{\mu}}$

MHT-CET Weightage: 6%

Sound Waves & Intensity

Sound Wave Properties:

- Longitudinal mechanical waves (travel in solids, liquids, gases).

$$v = \sqrt{\frac{B}{\rho}}$$

$v \to$ Speed, $B \to$ Bulk Modulus (B), $\rho \to$ Density (ρ)

Displacement: $s = s_m \cos(kx - \omega t)$

Pressure Change: $\Delta p = \Delta p_m \sin(kx - \omega t)$

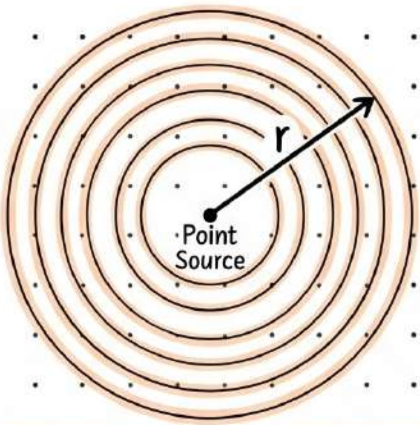
Pressure Amplitude: $\boxed{\Delta p_m = (v\rho\omega)s_m}$

Sound Intensity:

Intensity $I = \frac{1}{2}\rho v \omega^2 s_m^2$

Point Source: $I = \frac{P}{4\pi r^2}$

$P \to$ Power (P), $r \to$ Distance (r)



Sound Level in Decibels:

$$\boxed{\text{Sound Level } \beta = (10\text{ dB})\log_{10}\frac{I}{I_0}}$$

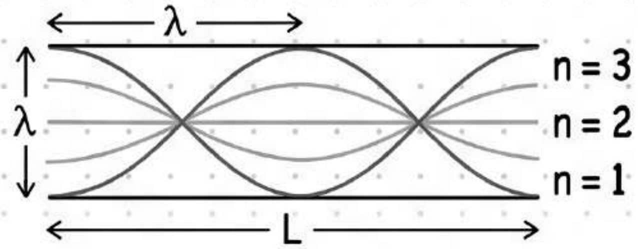
Reference Intensity: $I_0 = 10^{-12}\ \text{W/m}^2$

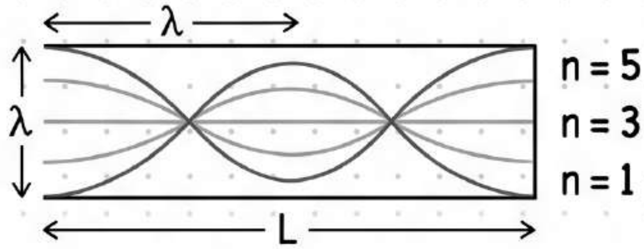
Every factor-of-10 increase in I adds +10 dB to β

MHT-CET Weightage: 6%

Pipes, Beats & Doppler Effect

Standing Wave in Open and Closed Pipes





A pipe open at both ends will resonate at frequencies

$$f = \frac{v}{\lambda} = \frac{nv}{2L}$$, where n = 1, 2, 3, ...

For a pipe closed at one end and open at the other, the resonant frequencies are

$$f = \frac{v}{\lambda} = \frac{nv}{4L}$$, where n = 1, 3, 5, ...

Beats

Beats arise when two waves having slightly different frequencies, f_1 and f_2, are detected together.

The beat frequency is $f_{beat} = |f_1 - f_2|$.

Doppler Effect

Doppler effect is a change in the observed frequency of a wave when the source or the detector moves relative to the transmitting medium.

The observed frequency [f'] is given in terms of the source frequency [f] by

$$f' = f\left(\frac{v \pm v_o}{v \pm v_s}\right).$$

v_o is the speed of the detector relative to the medium, v_s is that of the source, and v is the speed of sound in the medium.

The signs are chosen such that f' tends to be greater for motion toward and less for motion away.